

TOPIC 14/60

What are Embeddings?

Computers don't speak English.

How AI translates "meaning" into pure math.

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THE TRANSLATION PROBLEM

Neural networks can only process numbers. So, how do we teach an AI what a "Dog" is?

The Naive Approach:

Assign a random number to every word. (e.g., Apple=1, Dog=2, Cat=3, Car=4).

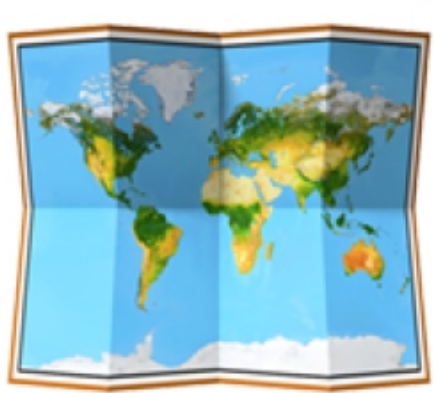
> Why it fails: "Dog (2)" and "Cat (3)" are mathematically close, but "Car (4)" is also close to "Cat". This completely destroys semantic relationships.

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THE 2D MAP ANALOGY



Plotting Meaning

Imagine a 2D graph. X-axis = "Fluffiness". Y-axis = "Size".

🐕 Dog: [0.8 Fluff, 0.5 Size]

🐈 Cat: [0.9 Fluff, 0.3 Size]

🚗 Car: [0.0 Fluff, 0.9 Size]

Now, Dog and Cat are physically close to each other on the map, while Car is far away. **This is an embedding.**

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SCALING IT UP

The Meaning Universe

Two dimensions (Fluffiness and Size) aren't nearly enough to capture the complexity of human language.

Hundreds of Dimensions

Modern LLMs use massive multidimensional spaces. OpenAI's embeddings use exactly **1,536 dimensions**.

Every word is plotted on 1,536 different conceptual axes simultaneously (gender, tone, color, emotion, etc).

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UNDER THE HOOD

The Vector

A Mathematical Barcode

An embedding is ultimately just a "Vector"—a very long list of decimal coordinates that pinpoint a concept's exact location in this universe.

```
"Apple" =  
[0.012, -0.841, 0.435, 0.112, -0.009, 0.992, 0.441, -0.321,  
0.884, ... and 1,527 more numbers]
```

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THE MAGIC TRICK

Distance = Similarity

Because concepts are plotted as physical coordinates, the AI can simply measure the distance between them.

Vector Math:

If you ask the AI to find synonyms for "Happy", it just looks at the embedding space and grabs the words that are located physically closest to it (like "Joyful" or "Glad").

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CONCEPT MATH

THE FAMOUS EXAMPLE

King - Man + Woman = Queen

Because meaning is mapped to coordinates, you can add and subtract concepts like math equations. The resulting coordinates perfectly land on the word "Queen".

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SCALING UP

Not just words

We don't just embed individual words. We can embed entire paragraphs, documents, images, and audio clips into the exact same space.

This means an image of a dog and the word "Dog" will share the exact same neighborhood in the vector space. This is how multimodal AI understands vision and text simultaneously.

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THE FOUNDATION OF AI

Semantic Search

When you Google a question, it embeds your question and finds documents with similar vector coordinates, even if they don't share exact keywords.

RAG Systems

Retrieval-Augmented Generation relies entirely on comparing the embeddings of user questions to a database of company documents.

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Embeddings

CONCEPT UNLOCKED

-  A coordinate system for meaning
-  Distance = Similarity
-  Allows math on human concepts
-  Powers modern search and RAG

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